

Traffic Signal Project

Arduino Uno + Tinkercad — Red / Yellow / Green LED Sequencer

Components Used

Component	Quantity
Arduino Uno R3	1
LED - Red	1
LED - Yellow	1
LED - Green	1
Resistor - 220 ohm	3
Breadboard	1
Jumper wires	as needed

Build Steps

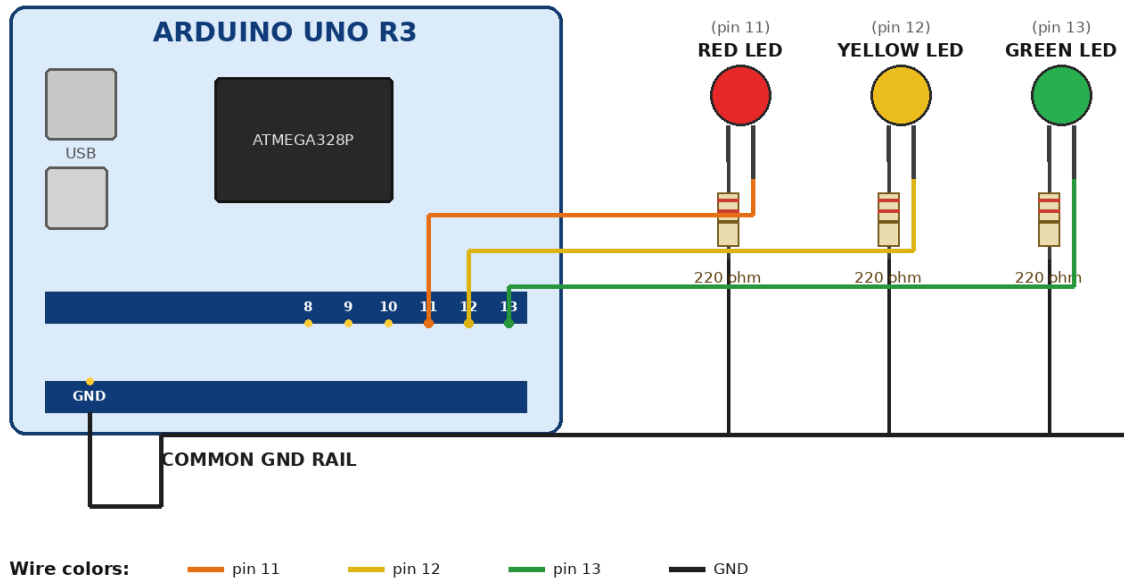
- 1 Open Tinkercad**
Go to tinkercad.com, log in, and click "Create new Circuit" to open a blank workspace.
- 2 Add the Arduino Uno R3**
Search "Arduino Uno R3" in the components panel and drag it onto the canvas.
- 3 Add a Breadboard**
Search "Breadboard" and drag it near the Arduino to make wiring easier and neater.
- 4 Add 3 LEDs**
Drag 3 LEDs onto the breadboard. Set their colors to Red, Yellow, and Green respectively (click each LED, then change its color property).
- 5 Add 3 Resistors (220 ohm each)**
Drag 3 resistors onto the breadboard, one next to each LED's cathode (short leg). Click each resistor and set its value to 220 ohm.
- 6 Wire the cathode side (LED to Resistor to GND)**
Connect each LED's cathode (short leg) to one end of its resistor. Connect the other end of each resistor to the breadboard's GND rail (blue line). Then connect the GND rail to Arduino's GND pin using one black jumper wire.
- 7 Wire the anode side (LED to Arduino pins)**
Connect Red LED anode to pin 11, Yellow LED anode to pin 12, and Green LED anode to pin 13, using jumper wires — see the diagram below for exact routing and colors.
- 8 Add the code**
Click the "Code" button on the Arduino, switch to "Text" mode, delete the placeholder code, and paste the sketch given below.

9 Run the simulation

Click "Start Simulation". The LEDs will cycle: Red (5s) -> Yellow (4s) -> Green (3s) -> repeat.

Circuit Diagram

White-background reference diagram — match these exact pin and wire connections in Tinkercad.



Wiring Summary

From	To	Wire Color
Red LED anode	Arduino pin 11	Orange
Yellow LED anode	Arduino pin 12	Yellow
Green LED anode	Arduino pin 13	Green
Red LED cathode	220 ohm resistor -> GND rail	Black
Yellow LED cathode	220 ohm resistor -> GND rail	Black
Green LED cathode	220 ohm resistor -> GND rail	Black
Breadboard GND rail	Arduino GND pin	Black

Full Arduino Code

```
void setup()
{
  pinMode(11, OUTPUT); // Red
  pinMode(12, OUTPUT); // Yellow
  pinMode(13, OUTPUT); // Green
}

void loop()
{
  digitalWrite(11, HIGH);
  digitalWrite(12, LOW);
  digitalWrite(13, LOW);
  delay(5000); // Red for 5 sec

  digitalWrite(11, LOW);
  digitalWrite(12, HIGH);
  digitalWrite(13, LOW);
  delay(4000); // Yellow for 4 sec

  digitalWrite(11, LOW);
  digitalWrite(12, LOW);
  digitalWrite(13, HIGH);
  delay(3000); // Green for 3 sec
}
```

Tinkercad Circuit Screenshot

